

LAB 3 REPORT

Component Modelling & Architectural Pattern Selection

Project: Peer Skill Exchange & Mentorship Network

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Selected Architecture: Layered Architecture

1. Objective

To evaluate suitable architectural styles for the Peer Skill Exchange & Mentorship Network, select an appropriate architecture, and model the system using a UML Component Diagram showing major components, interfaces, dependencies, and interactions.

2. Project Context

The Peer Skill Exchange & Mentorship Network is a time-banking platform where members exchange skills. It supports skill profiles, mentor discovery, session scheduling, mutual completion confirmation, time-credit transfer, ratings and feedback, and calendar integration.

3. Architectural Style Analysis

Architecture	Structure	Advantages	Limitations
Layered Architecture	Presentation, business, and data/integration layers.	Clear separation of concerns; easy to understand and maintain.	Can introduce communication overhead.
Microservices Architecture	Independent small services.	Independent deployment, scaling, and fault isolation.	Higher operational and distributed-system complexity.
Client-Server Architecture	Clients communicate with a centralized server.	Centralized control and simple deployment.	Possible bottleneck and single point of failure.

4. Architecture Selection

Layered Architecture is selected because it clearly separates user-facing presentation functions, business services implementing the project rules, and data/external integration responsibilities. It provides sufficient modularity for the current project without the additional deployment and distributed-system complexity of Microservices.

5. Components

Component	Layer	Responsibility
Member Portal / UI	Presentation	Learner and mentor interaction.
Profile Management	Business	Manages skills, expertise, goals, and interests.
Mentor Discovery	Business	Searches and matches suitable mentors.
Session Management	Business	Handles booking, completion, and session status.
Time-Credit Service	Business	Validates and transfers time credits.
Feedback & Rating Service	Business	Handles ratings and feedback after verification.
Data Repository	Data	Stores profiles, sessions, credits, and feedback.
Calendar Integration	External	Supports iCal calendar export/synchronization.

6. UML Component Diagram

The final UML Component Diagram is included on the following page.

7. Interfaces and Interactions

Components	Interface	Purpose
Member Portal / UI ↔ Profile Management	IProfileService	Profile API
Member Portal / UI ↔ Mentor Discovery	ISearchService	Search API
Member Portal / UI ↔ Session Management	ISessionService	Booking / Confirmation API
Session Management ↔ Time-Credit Service	ICreditService	Credit Validation & Update
Session Management ↔ Feedback & Rating Service	IFeedbackService	Verified Session API
Session Management ↔ Calendar Integration	ICalendarService	iCal Export / Sync

8. Security Considerations

Sensitive operations are handled by dedicated business services rather than directly by the user interface. Mutual completion confirmation and verified session status control time-credit transfer and feedback eligibility. Service boundaries also provide clear points for validation and authorization.

9. Performance Considerations

Focused components reduce unnecessary processing and keep responsibilities separate. Centralized data access avoids duplicated persistence logic, while calendar integration is isolated from core session processing.

10. Final Justification

Architecture Choice: Layered Architecture was chosen for the Peer Skill Exchange & Mentorship Network.

Reason 1: It clearly separates presentation, business logic, and data/integration responsibilities, making the system easier to understand and maintain.

Reason 2: It suits the current project scope and provides modularity without the additional operational complexity of a Microservices architecture.

Security Advantage: Business services consistently control validation, verified completion, credit updates, and feedback eligibility.

Performance Benefit: The separation of responsibilities keeps operations focused and centralizes data access, reducing unnecessary duplication.

11. Conclusion

The Layered Architecture provides a modular and understandable structure for the Peer Skill Exchange & Mentorship Network. The component model supports the major requirements of profiles, mentor discovery, session management, time credits, feedback, data persistence, and calendar integration.

Peer Skill Exchange & Mentorship Network – UML Component Diagram

